Abhishek Rithik Origanti

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SUMMARY

Data Engineer with an M.S. in Data Science and 3+ years building pipelines for healthcare and financial analytics. Experienced in Python, SQL, PySpark, AWS, Databricks, and Snowflake, from ingestion and modeling to data quality and production support. Reduced healthcare batch runtime by 32% and processed 20M+ lending records. Built a Bedrock code-review assistant, used Claude and Cursor to test changes, and developed GPT-4 search to help analysts find information faster.

WORK EXPERIENCE

Optum *Remote, US*

Data Engineer *Sep 2025 – Present*

- Engineered PySpark and AWS Glue pipelines for multi-terabyte healthcare claims in S3, using EMR partitioning and Glue Data Catalog metadata to cut recurring batch runtime by 32% while verifying record totals against source extracts.
- Modeled claims and eligibility marts in Snowflake using SQL and dbt incremental loads orchestrated via Airflow, reconciling member and provider keys across feeds so operations teams could run recurring reports 35% faster.
- Built reusable claim-level feature tables in Python and Spark for payment-integrity models, working with data scientists on time-based training splits and null-handling rules to support 4 production fraud and overpayment experiments.
- Developed a Python and Amazon Bedrock PR assistant that checks SQL/PLSQL changes against team quality rules and posts cited review comments in GitHub Actions, reducing manual database code review effort by roughly 60%.
- Debugged complex SQL transformation failures and expanded pytest coverage using Cursor and Claude on masked claim samples, catching upstream schema drift early and reducing recurring source-to-target reconciliation tickets by 25%.
- Automated AWS and Snowflake environment promotions with Terraform and GitHub Actions, separating dev and prod configs with documented rollback steps to cut manual release tasks in half across 8 scheduled pipelines.
- Troubleshot late-arriving files, task failures, and SLA delays across Airflow and Glue workflows using CloudWatch, EventBridge, and Athena SQL, documenting root-cause runbooks that trimmed average on-call recovery time from 90 to 45 minutes.

Bajaj Finserv *Bengaluru, India*

Data Engineer – AI/ML *Jan 2023 – Jul 2024*

- Designed Azure Databricks and Data Factory pipelines for consumer-loan and repayment data, applying PySpark, SQL, and Delta Lake incremental loads on ADLS Gen2 to process 20M+ records and cut monthly portfolio refresh runtime by 30%.
- Created reusable borrower and repayment feature sets by joining bureau, loan, and payment histories in PySpark, aligning observation windows with credit-risk analysts to cut recurring feature-prep work by 28% before model runs.
- Tracked credit-risk model baselines using scikit-learn and MLflow across time-based validation splits, logging feature versions and class-imbalance metrics across 15+ experiments to prevent target leakage prior to deployment.
- Piloted an internal RAG search service in 2024 using Python, Azure OpenAI GPT-4, vector search, and FastAPI over lending policy docs, verifying responses against cited passages to drop analyst lookup time from 12 to under 4 minutes.
- Implemented schema, freshness, and row-count reconciliation checks on Delta Lake tables, catching incomplete bureau and payment loads upstream before corrupted records could break risk models or daily campaign extracts.
- Partnered with risk and reporting teams to standardize delinquency, repayment, and campaign definitions, publishing governed SQL views and Power BI datasets that eliminated repetitive join logic across 20+ weekly business reports.

Tech Mahindra *Hyderabad, India*

SQL Developer Intern *May 2022 – Dec 2022*

- Assisted senior developers with SSIS and Azure Data Factory loads from SQL Server, Salesforce, APIs, and flat files into reporting tables, adding file validation and retry logic that shortened the overnight batch window by 25%.
- Tuned T-SQL queries, stored procedures, and non-clustered indexes under senior DBA guidance, analyzing execution plans and verifying row counts to cut heavy query runtimes by up to 40% and report runs from 50 to 32 minutes.
- Built star-schema Power BI datasets for sales operations with reusable dimensions, DAX, and Power Query; helped business analysts identify $190K+ in potential operational inefficiencies while shortening refresh cycles by 30%.

EDUCATION

University of Maryland *College Park, MD*

Master of Science in Data Science – GPA: 3.57/4.0 *Aug 2024 – May 2026*

SKILLS AND PROFICIENCY

Programming & Databases: Python, SQL, T-SQL, PL/SQL, Pandas, PostgreSQL, SQL Server, Snowflake, Query Optimization
Data Processing & Engineering: Apache Spark (PySpark, Spark SQL), Databricks, Delta Lake, Parquet, dbt, AWS Glue, SSIS, ETL/ELT, Dimensional Modeling
Cloud & Orchestration: AWS (S3, EMR, Athena, CloudWatch, EventBridge), Azure (ADLS Gen2, Data Factory), Apache Airflow
DataOps & Quality: Terraform, Docker, Git, GitHub Actions, CI/CD, pytest, dbt Tests, Reconciliation, Data Lineage
AI & Analytics: Amazon Bedrock, Azure OpenAI (GPT-4), Claude, Cursor, RAG, scikit-learn, MLflow, FastAPI, Power BI, Tableau

CERTIFICATIONS

- Databricks Certified Data Engineer Professional | *Verify*
- Microsoft Certified: Power BI Data Analyst Associate | *Verify*
- AWS Certified Solutions Architect – Associate | *Verify*